

Spaceship Titanic Classification Project



milkloong @MLW



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01

Project Overview



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Project Introduction



Competition Information

- Kaggle tabular binary classification task
- Goal: predict whether each passenger was transported
- Input: passenger, travel, cabin, and spending information
- Target: Transported = True / False
- Evaluation metric: classification accuracy

Team Information

Team: milkloong @MLW

We built a complete machine learning workflow, including EDA, preprocessing, feature engineering, five model implementations, model comparison, and Kaggle submission tracking.



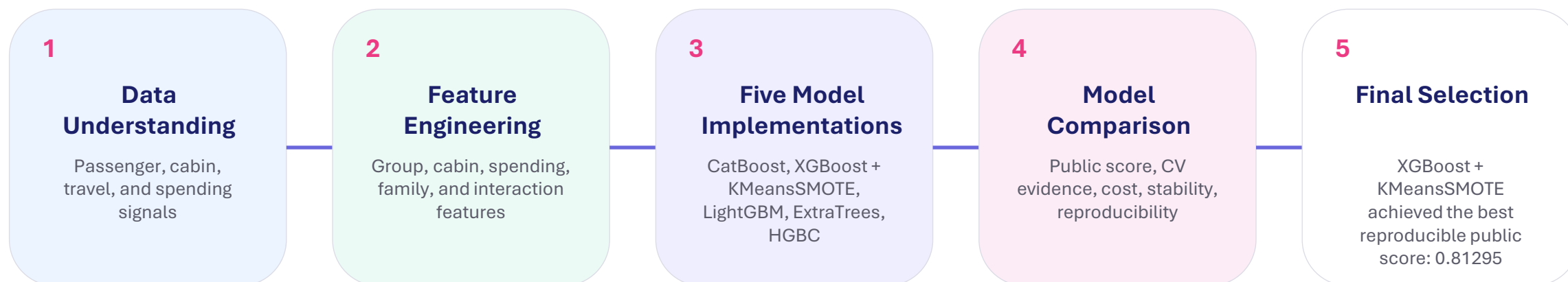
Final Result

- Implemented models:
 - CatBoost
 - XGBoost + KMeansSMOTE
 - LightGBM
 - ExtraTrees
 - HGBC
- Best model: **XGBoost + KMeansSMOTE**
- Best public Kaggle score: **0.81295**





Core Storyline



The project did not improve in a straight line. Intermediate models were still valuable because they guided the final model selection.

02

Dataset and Feature Engineering



Hou Rui

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Feature Engineering Overview

PassengerId

- GroupID, GroupMember, GroupSize, IsSolo
- Purpose: Capture group travel pattern

Cabin

- Deck, CabinNum, Side, Deck_Side
- Purpose: Capture cabin location

Spending

- TotalSpend, NoSpend, LuxurySpend, LogTotalSpend
- Purpose: Capture passenger activity

Name

- Surname, SurnameSize
- Purpose: Capture family signals

Interactions

- Cryo_NoSpend, HomePlanet_Deck, Deck_Side
- Purpose: Capture combined effects

Goal: convert raw identifiers, strings, and sparse spending records into structured signals for the models.

Feature Engineering Pipeline



01 Raw Dataset

PassengerId, Cabin, Name, spending columns, CryoSleep, HomePlanet, Destination



02 Feature Engineering Operations

ID parsing, cabin splitting, spending aggregation, name extraction, interaction creation



03 Engineered Features

GroupID / GroupSize / IsSolo; Deck / CabinNum / Side; TotalSpend / NoSpend / LogTotalSpend; Surname / SurnameSize; Cryo_NoSpend / HomePlanet_Deck / Deck_Side



04 Model-ready Input

Features used by CatBoost, XGBoost, LightGBM, and HGBC



Feature Engineering Purpose

Group Patterns

Passengers in the same group may share similar travel behavior. PassengerId helped us recover group-level structure.

Spending Behavior

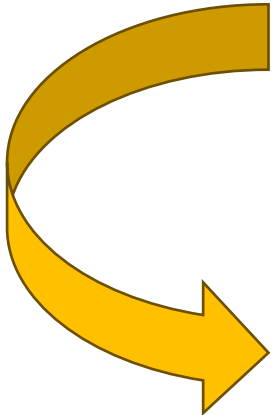
CryoSleep and spending are strongly connected. NoSpend and TotalSpend helped represent passenger activity.

Cabin Location

Cabin deck and side may reflect spatial patterns related to transportation.

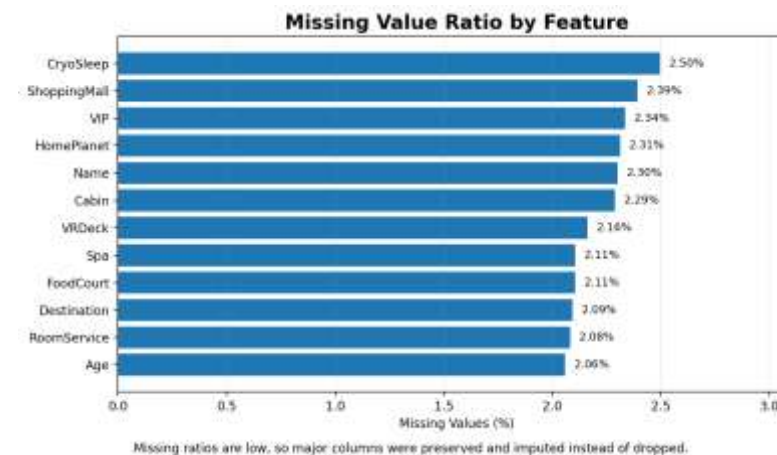
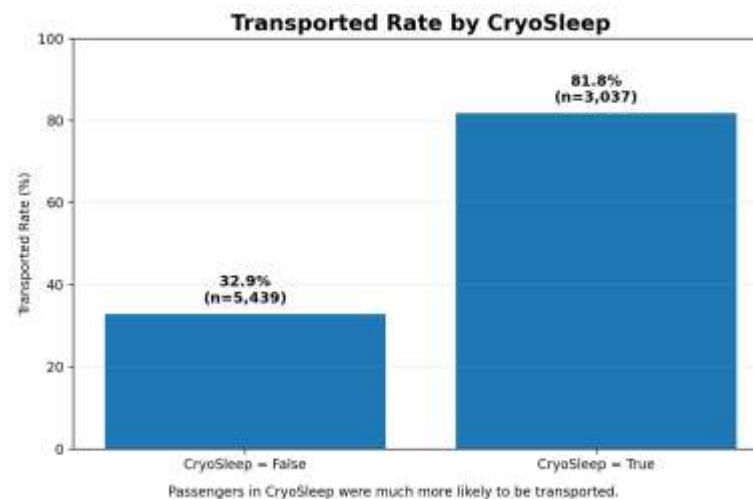
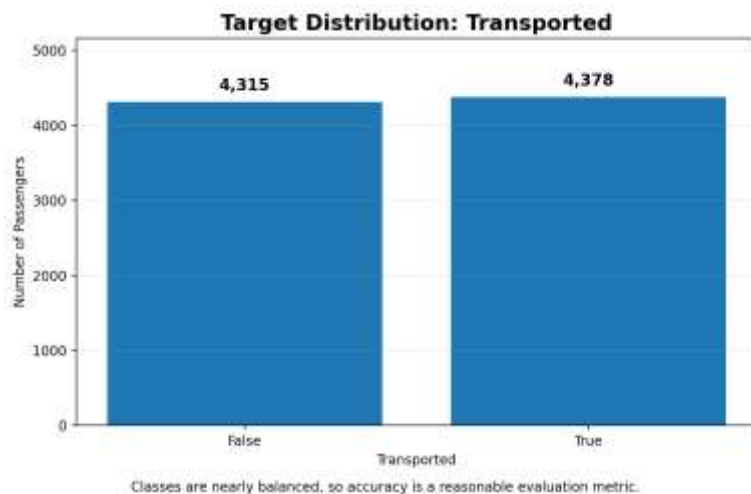
Categorical Signals

HomePlanet, Destination, Deck, Side, and Surname contain important categorical information. These categorical signals were important for later model



Raw Features	
1	PassengerId
2	Cabin
3	Name
4	Spending Columns (RoomService, FoodCourt, ShoppingMall, Spa, VRDeck)
5	Categorical / Context Columns (HomePlanet, CryoSleep, Destination, VIP)
Useful Signals	
GroupID, GroupSize, GroupMember, IsSolo	
Deck, CabinNum, Side, Deck_Side	
Surname, SurnameSize	
TotalSpend, NoSpend, LuxurySpend, BasicSpend, LogTotalSpend	
Cryo_NoSpend, HomePlanet_Deck, important category features for CatBoost	

EDA Findings Used for Feature Design



Balanced Target

Accuracy is meaningful because the two target classes are nearly balanced.

Strong CryoSleep Signal

CryoSleep motivated features such as NoSpend and Cryo_NoSpend.

Moderate Missingness

Missing ratios are low, so major columns were preserved and imputed.

03

Model Implementations



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Overall Model Strategy



CB

CatBoost

Strong experimental route

- Native categorical handling + feature blending

Public score: 0.80967

XG

XGBoost + KMeansSMOTE

Final selected model

- Compact features + resampling

Public score: 0.81295

LGB

LightGBM

Fast boosting benchmark

- Efficient feature / parameter comparison

Public score: 0.80734

HGB

HGBC

Reproducible baseline

- Simple scikit-learn reference route

Public score: 0.80266

ET

ExtraTrees

High-variance ensemble benchmark

- Additional tree-ensemble comparison

Public score: 0.80313

Score

**Categorical
suitability**

Stability

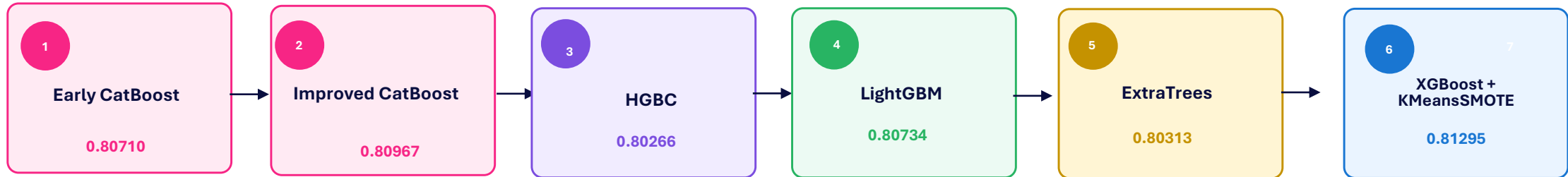
Reproducibility

Final Decision

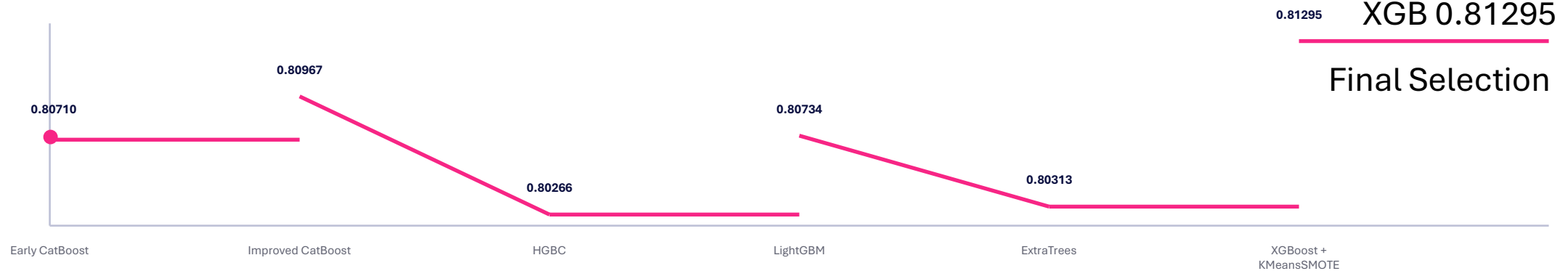
Strongest reproducible public score generated the final submission reliably.

Experimental Mainline: Real Model Order

We did not skip intermediate attempts. The final model was selected after comparing lower-scoring and higher-scoring routes.



Kaggle Public Score Progression



Key message: the score did not increase in a straight line; the final decision prioritized reproducibility and a strong public score.

CatBoost Route: Early and Improved Experiments



Evolution

Early CatBoost scored 0.80710. After richer feature engineering, Improved CatBoost reached 0.80967. This route proved CatBoost was strong, but it was not used as the final reproducible pipeline.



Feature Engineering

Used group, cabin, surname / family, spending, CryoSleep consistency, and missing-indicator features.



Validation

Used cross-validation ideas and categorical-feature handling to test whether CatBoost fit the engineered data representation.



Role

CatBoost handled categorical variables naturally and informed our feature design, but the final selected reproducible pipeline was XGBoost + KMeansSMOTE.

Workflow



CatBoost Route Evidence



Improved Score
0.80967

Earlier CatBoost:

0.80710

Improved CatBoost:

0.80967

Important experimental route

HGBC Model: Scikit-Learn Baseline

Feature Building

Used a custom feature builder for passenger, cabin, surname, spending, and interaction features.

Preprocessing

SimpleImputer handled missing values, and OrdinalEncoder converted categorical variables.

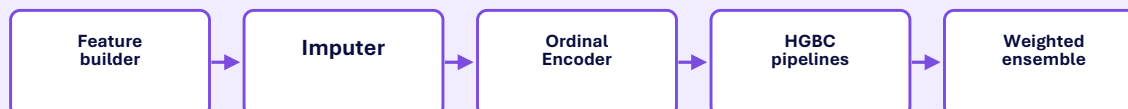
Ensemble

A weighted ensemble of multiple HistGradientBoostingClassifier pipelines was used.

Role

HGBC provided a simple, reproducible scikit-learn baseline for evaluating engineered features.

Workflow



Real Output: HGBC Submission



Public Score

0.80266

Test predictions:

2,554 True / 1,723 False

Reproducible reference model

Fast Benchmark Model: LightGBM

Feature Engineering

Used group, cabin, spending, age, surname, and missing-indicator features.

Preprocessing

Numerical features were imputed, while categorical features were one-hot encoded.

Validation

Used 10-fold StratifiedKFold, OOF probability prediction, threshold search, and test probability averaging.

Role

LightGBM served as a fast-boosting benchmark for comparing feature and preprocessing choices.

Workflow



Real Output: LightGBM Submission



Public Score

0.80734

Test predictions:

2,230 True / 2,047 False

Fast benchmark route

ExtraTrees Model: High-Variance Ensemble Benchmark

Feature Engineering

Used rich features including group, cabin, spending, age, surname, and missing-indicator features.

Preprocessing

Numerical features used median imputation, while categorical features used most-frequent imputation plus one-hot encoding.

Validation

Used 12-fold StratifiedKFold, OOF probability prediction, threshold search, and test probability averaging.

Role

ExtraTrees served as a high-variance ensemble benchmark for comparing feature and preprocessing choices.

Workflow



Real Output: ExtraTrees Submission



Public Score

0.80313

Test predictions:

2,258 True / 2,019 False

Additional ensemble benchmark

XGBoost + KMeansSMOTE Model: Strongest Comparison Route

DB

Feature Engineering

Used a compact feature set focused on CryoSleep, spending consistency, passenger group signals, and cabin-side extraction.



Preprocessing

Categorical variables were one-hot encoded and numerical features were prepared for a compact XGBoost pipeline.



Resampling

KMeansSMOTE was applied to address class imbalance by generating synthetic samples within local clusters.



Role

This became our final reproducible model because it reran successfully and achieved the strongest reproducible public score.

Workflow



Real Output: XGBoost + KMeansSMOTE Submission



Public Score

0.81295

Final reproducible model

Public score:

0.81295

Selected final reproducible pipeline

04

Model Comparison and Selection



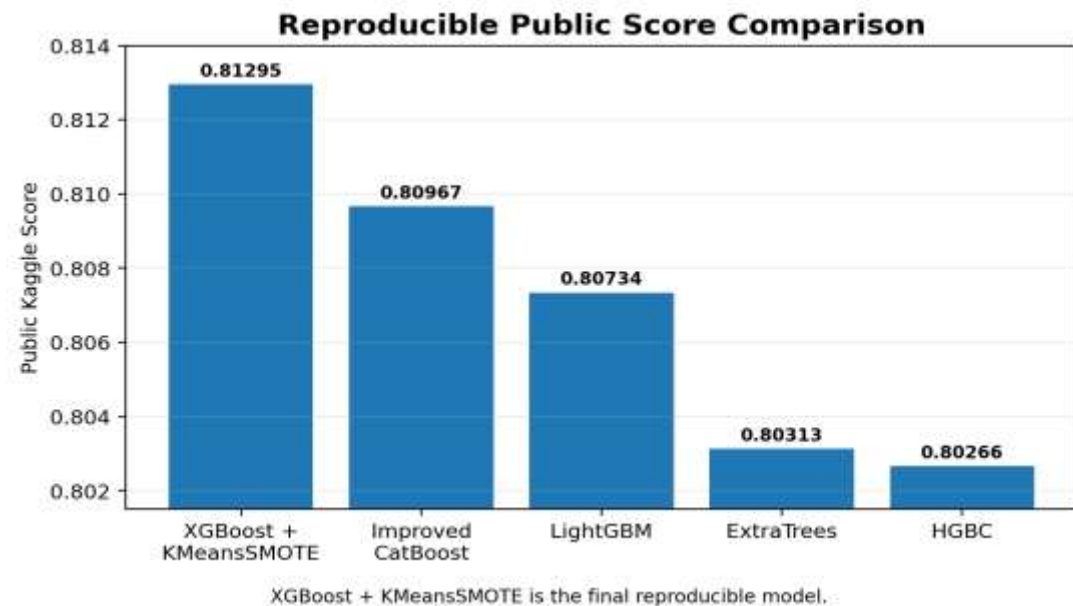
Guo Yitao

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Model Comparison Results

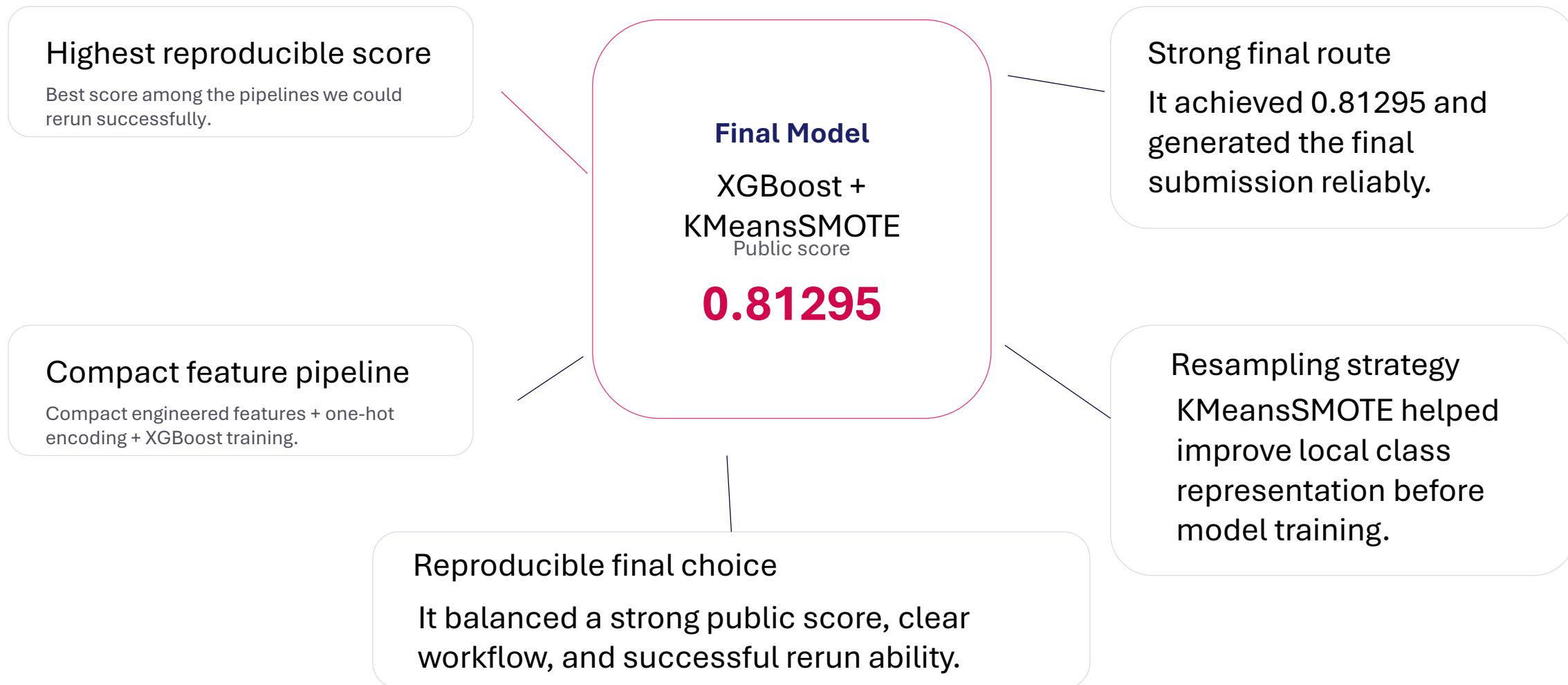
Rank	Model	Public score	Role
1	XGBoost + KMeansSMOTE	0.81295	Final reproducible model
2	Improved CatBoost	0.80967	Strong CatBoost route
3	LightGBM	0.80734	Fast benchmark
4	ExtraTrees	0.80313	Additional tree-based comparison
5	HGBC	0.80266	Reproducible baseline



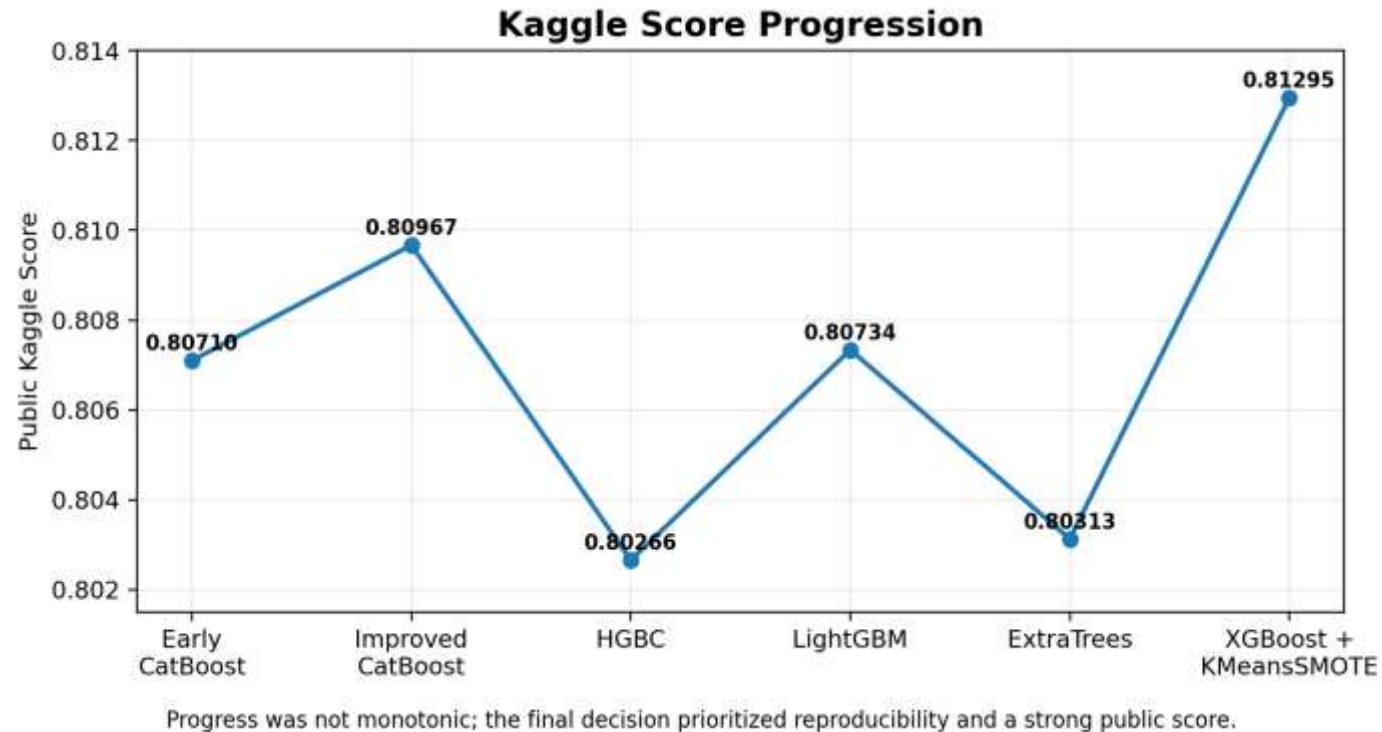
Key takeaway: XGBoost + KMeansSMOTE is the final reproducible model; CatBoost remains a strong experimental route.



Why XGBoost + KMeansSMOTE Was Selected



Kaggle Score Progression



Interpretation

- The project did not improve in a straight line.
- HGBC, LightGBM, and ExtraTrees provided comparison evidence.
- XGBoost + KMeansSMOTE achieved the best reproducible final score.
- The final decision prioritized both score and reproducibility.

Final improvement

From 0.80710 to 0.81295
Total gain: +0.00585

05

Challenges and Lessons Learned



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What we learned ?

Rank	Model	Public score
1	XGBoost + KMeansS MOTE	0.81295
2	Improved CatBoost	0.80967
3	LightGBM	0.80734
4	ExtraTrees	0.80313
5	HGBC	0.80266

Why XGBoost+KMeansSMOTE can get the highest score?

● Challenge

Passenger behavior was not uniform across all groups.

● Risk

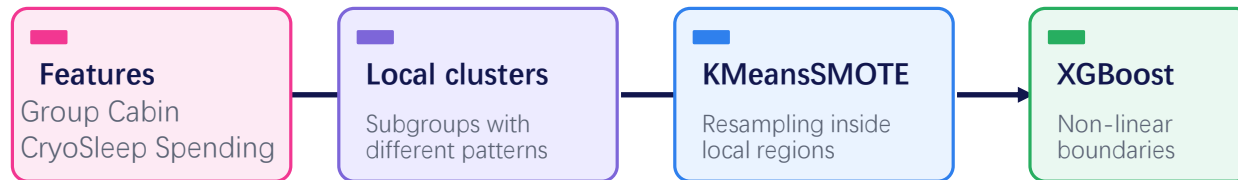
A single global pattern may miss subgroup-level signals.

Because Spaceship Titanic is not a simple problem;
(A feature can not determine the final prediction results; And there are local subgroup differences in the data)

What we learned ?

Different passenger subgroups may follow different transportation patterns.

How the final model handles local patterns



Local subgroup examples



Final selected model
XGBoost + KMeansSMOTE
Public score: 0.81295

Core idea: instead of assuming one global rule, the final model learns subgroup-level relationships from engineered features.

Solution

KMeansSMOTE resampled data within local clusters before XGBoost training.

Result

XGBoost + KMeansSMOTE achieved the best public score: 0.81295.

Lesson: Local subgroup patterns matter; the final model should capture both feature interactions and local data structure.



Challenge : Model-Specific Preprocessing

Model	Categorical strategy	Preprocessing focus	Reason
CatBoost	Native categorical handling	Preserve category columns	Matches categorical-heavy engineered features
XGBoost + KMeansSMOTE	One-hot / encoded features	Compact features + resampling	Requires numerical input
LightGBM	One-hot / encoded features	Rich features + 10-fold OOF	Fast benchmark
ExtraTrees	One-hot / encoded features	Rich features + 12-fold CV	Tree ensemble comparison
HGBC	Ordinal encoding	SimpleImputer + OrdinalEncoder	Reproducible scikit-learn baseline

Impact

The same raw columns could behave differently depending on model compatibility.

Solution

We built model-specific pipelines instead of forcing identical preprocessing.

Lesson

Preprocessing should match each model's assumptions and strengths.

06

Conclusion and Team Contribution



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Project Conclusion and Team Contribution

Final Selected Model **XGBoost + KMeansSMOTE pipeline**

Best Public Score

0.81295

Accuracy

Project Conclusion

- Built a complete ML pipeline: EDA → Feature Engineering → 5 Models → Kaggle submission
- Final model: XGBoost + KMeansSMOTE — best public score 0.81295
- Performance ceiling was set at the feature engineering stage, not model selection
- The right model for your feature design matters more than model complexity

Team Contribution

- Guo Yitao: CatBoost route analysis, GitHub organization, report/PPT integration.
- Yang Zhongxing: XGBoost + KMeansSMOTE final modeling and comparison experiments.
- Hou Rui: LightGBM benchmark model and validation experiments.
- Chen Ziheng: HGBC baseline and supporting analysis.
- Lu Kejie: ExtraTrees model implementation and comparison analysis.
- Whole team: EDA, feature engineering discussion, final review, and presentation preparation.

Final takeaway: the selected result came from meaningful feature engineering, model comparison, and prioritizing reproducibility under project constraints.

Thank you!



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